Ex - 08	Data Visualization Using Tableau

AIM

To visualize data using Tableau by connecting to various data sources, creating bar charts, line charts, and pie charts, implementing calculated fields, and building interactive dashboards and stories.

PROBLEM STATEMENT

To analyze and visualize data using Tableau by connecting Excel, CSV, and SQL databases, creating basic charts, performing calculations, and developing interactive dashboards to present meaningful insights.

PROCEDURE

1. Open Tableau Desktop and explore the interface, including the Data Source page, worksheet, Rows shelf, Columns shelf, Marks card, and Show Me panel.
2. Connect to an Excel or CSV dataset by selecting the appropriate option under Connect, choosing the file, and loading the required data.
3. Connect to a SQL database by selecting the required database connector, entering the server details, providing authentication if needed, and selecting the required tables.
4. Check the imported data, verify the data types, rename fields if necessary, and remove unwanted columns or records to prepare the dataset for visualization.
5. Create a bar chart by dragging a categorical field to Columns and a numerical field to Rows, then select Bar Chart from the Show Me panel.
6. Create a line chart by dragging a date field to Columns and a numerical field to Rows, then select Line from the Marks card to display trends over time.
7. Create a pie chart by selecting Pie from the Marks card, dragging a category field to Color, and dragging a numerical field to Angle to display category-wise proportions.

8. Create a calculated field by selecting Analysis → Create Calculated Field, entering a formula, such as [Sales] * [Quantity], and clicking OK to generate a new calculated field.
9. Build a dashboard by selecting New Dashboard, dragging the required worksheets into the dashboard, and arranging charts, titles, legends, and filters to create an interactive layout.
10. Create a story by selecting New Story, adding worksheets or dashboards as story points, providing captions, and saving the workbook for further analysis and presentation.

IMPLEMENTATION

1. Calculated Field – Total Amount

    [Sales] * [Quantity]

2. Calculated Field – Sales Category

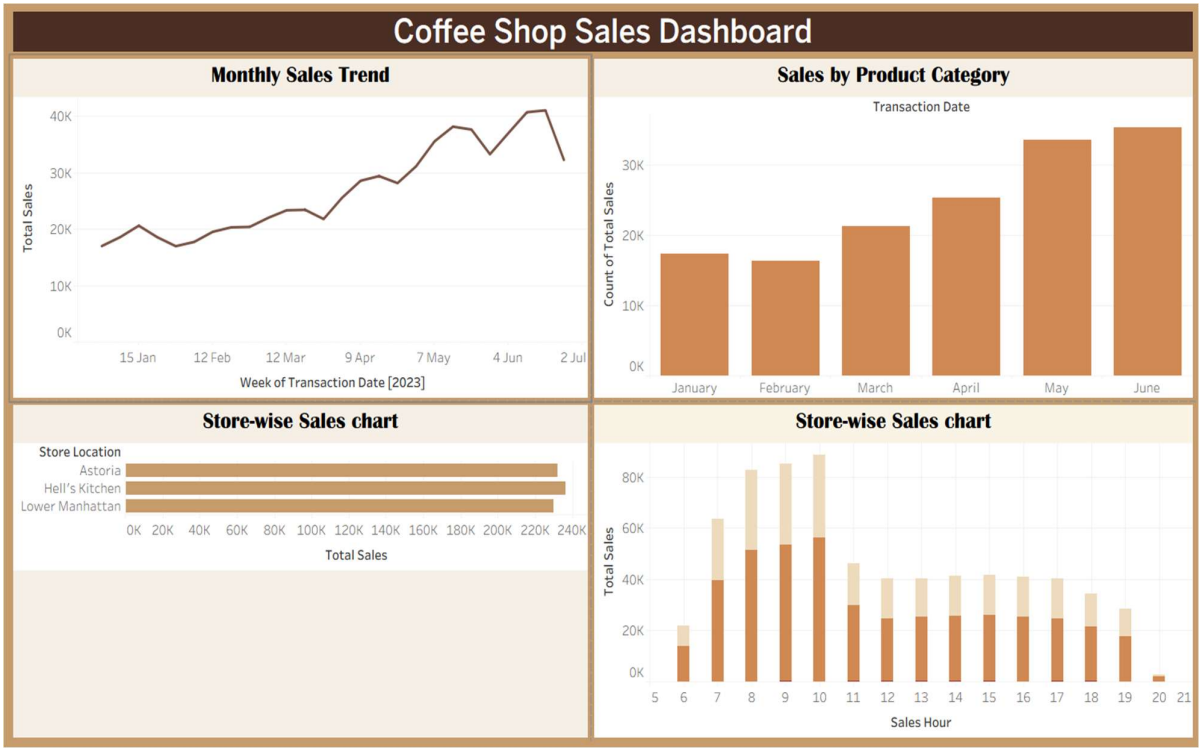
    IF [Sales] >= 1000 THEN "High Sales"

    ELSE "Low Sales"

    END

Dashboard components:

- Bar chart – Product-wise sales
- Line chart – Monthly sales trend
- Pie chart – Category-wise sales distribution
- Filter – Select a category or date range
- Story – Present visualizations in a sequence



RESULT

Data visualization using Tableau was successfully studied and implemented. Data was connected from different sources, bar charts, line charts, and pie charts were created, calculated fields were implemented, and interactive dashboards and stories were developed to analyze and present meaningful insights.